How to run cross product and vector rotation.py

Once you have installed Python, VPython, and NumPy, you are ready to run this code. *(Note: This code requires a tool called* `numpy`*. If you don't have it, open your computer's terminal, type* `pip install numpy`*, and press Enter before starting).*

Step 1: Open Python IDLE

Search for **IDLE** on your computer (this is the coding program that comes with Python) and open it.

Step 2: Open your file

1. In the top menu of IDLE, click **File** -> **Open**.
2. Find and select your `cross product and vector rotation.py` file.

Step 3: Run the code

When your code opens in a new window, press the **F5** key on your keyboard (or click **Run** -> **Run Module** in the top menu).

Step 4: Enter your vector values

A text window (the IDLE Shell) will ask you to enter the numbers for two vectors:

1. Type a number and press `Enter` for the X, Y, and Z components of **Vector A**.
2. Do the same for the X, Y, and Z components of **Vector B**.

Step 5: View the 3D Scene and Animation

A new web browser tab will open showing your 3D space:

- **Red Arrow:** Vector A
- **Green Arrow:** Vector B
- **Blue Arrow:** The Cross Product vector (A x B)
- You will also see a light blue plane (surface) between Vector A and Vector B.

CRITICAL STEP FOR ANIMATION: To see the mathematical rotation happen, you must **click anywhere on the 3D scene** with your mouse. Vector A will then smoothly rotate towards Vector B!

- **To rotate your view:** Right-click and drag your mouse.
- **To zoom in/out:** Use your mouse wheel.